RoomGuard V1.1 BASEBOARD — REVIEW DRAFT — NOT FOR FABRICATION

15 pins/side, pitch2.54 confirmed. Both ESP row spacings and module connector orders require measurement.

POWER / VIN / VREF

J1
5V INPUT 1=+5V 2=GND

RAW_5V	1	1:RAW_5V
GND	2	2:GND

F1
PTC hold1.1A / trip2.2A

RAW_5V	1	1:RAW_5V
FUSED_IN	2	2:FUSED_IN

Q1
AO3401A reverse-polarity

RP_GATE	1	1:RP_GATE
FUSED_OUT	2	2:FUSED_OUT
FUSED_IN	3	3:FUSED_IN

R1
100k

RP_GATE	1	1:RP_GATE
GND	2	2:GND

SW1
POWER SWITCH

FUSED_OUT	1	1:FUSED_OUT
PSV	2	2:PSV

C1
100uF/10V

PSV	1	1:PSV
GND	2	2:GND

C2
10uF

PSV	1	1:PSV
GND	2	2:GND

C3
100nF

PSV	1	1:PSV
GND	2	2:GND

U1
AP2112K-3.3

PSV	1	1:PSV
GND	2	2:GND
PSV	3	3:PSV
	X 4	4:NC_U1_4
SENSOR_3V3	5	5:SENSOR_3V3

C4
1uF

PSV	1	1:PSV
GND	2	2:GND

C5
1uF

SENSOR_3V3	1	1:SENSOR_3V3
GND	2	2:GND

C6
10uF

SENSOR_3V3	1	1:SENSOR_3V3
GND	2	2:GND

JP1
SELECT ESP VREF (3?:1-2 / 8266:2-3)

ESP32_3V3	1	1:ESP32_3V3
ESP_3V3	2	2:ESP_3V3
ESP8266_3V3	3	3:ESP8266_3V3

JP2
ESP32 VIN ENABLE

PSV	1	1:PSV
ESP32_VIN	2	2:ESP32_VIN

JP3
ESP8266 VIN ENABLE

PSV	1	1:PSV
ESP8266_VIN	2	2:ESP8266_VIN

ANALOG BUFFER / DIVIDERS

U2
MCP6002-I/P unity buffer

MQ_BUF	1	1:MQ_BUF
MQ_BUF	2	2:MQ_BUF
MQ_AO	3	3:MQ_AO
GND	4	4:GND
GND	5	5:GND
OPB_OUT	6	6:OPB_OUT
OPB_OUT	7	7:OPB_OUT
PSV	8	8:PSV

R2
20k 1%

MQ_BUF	1	1:MQ_BUF
ADC32_DIV	2	2:ADC32_DIV

R4
1k

ADC32_DIV	1	1:ADC32_DIV
ADC32	2	2:ADC32

R5
56k 1%

MQ_BUF	1	1:MQ_BUF
ADC8266_DIV	2	2:ADC8266_DIV

R7
1k

ADC8266_DIV	1	1:ADC8266_DIV
ADC8266	2	2:ADC8266

C7
100nF

PSV	1	1:PSV
GND	2	2:GND

R3
10k 1%

ADC32_DIV	1	1:ADC32_DIV
GND	2	2:GND

C8
10nF

ADC32	1	1:ADC32
GND	2	2:GND

R6
10k 1%

ADC8266_DIV	1	1:ADC8266_DIV
GND	2	2:GND

C9
10nF

ADC8266	1	1:ADC8266
GND	2	2:GND

DIGITAL LEVEL ADAPTATION

U3
SN74LVC2G17DCKR

SMOKE_IN	1	1:SMOKE_IN
GND	2	2:GND
FLAME_IN	3	3:FLAME_IN
FLAME_MCU	4	4:FLAME_MCU
ESP_3V3	5	5:ESP_3V3
SMOKE_MCU	6	6:SMOKE_MCU

C10
100nF

ESP_3V3	1	1:ESP_3V3
GND	2	2:GND

R8
1k

MQ_DO	1	1:MQ_DO
SMOKE_IN	2	2:SMOKE_IN

R9
10k

MQ_DO	1	1:MQ_DO
PSV	2	2:PSV

R10
1k

FLAME_DO	1	1:FLAME_DO
FLAME_IN	2	2:FLAME_IN

R11
10k

FLAME_DO	1	1:FLAME_DO
SENSOR_3V3	2	2:SENSOR_3V3

Q2
BSS138 DHT bidirectional

ESP_3V3	1	1:ESP_3V3
DHT_MCU	2	2:DHT_MCU
DHT_SENSOR	3	3:DHT_SENSOR

R12
10k

DHT_MCU	1	1:DHT_MCU
ESP_3V3	2	2:ESP_3V3

R13
10k

DHT_SENSOR	1	1:DHT_SENSOR
SENSOR_3V3	2	2:SENSOR_3V3

Q3
BSS138 buzzer low-active

ESP_3V3	1	1:ESP_3V3
BUZZ_MCU	2	2:BUZZ_MCU
BUZZ_INPUT	3	3:BUZZ_INPUT

R14
10k

BUZZ_MCU	1	1:BUZZ_MCU
ESP_3V3	2	2:ESP_3V3

R15
10k

BUZZ_INPUT	1	1:BUZZ_INPUT
PSV	2	2:PSV

R16
330

LED_R_MCU	1	1:LED_R_MCU
LED_R	2	2:LED_R

R17
330

LED_G_MCU	1	1:LED_G_MCU
LED_G	2	2:LED_G

R18
330

LED_B_MCU	1	1:LED_B_MCU
LED_B	2	2:LED_B

TWO SOCKETS — POPULATE ONE ESP ONLY

J7
J32L 15pin SOCKET

	1	1:NC32_EN
	2	2:NC32_VP
	3	3:NC32_VN
ADC32	4	4:ADC32
	5	5:NC32_35
	6	6:NC32_32
	7	7:NC32_33
FLAME_MCU	8	8:FLAME_MCU
SMOKE_MCU	9	9:SMOKE_MCU
DHT_MCU	10	10:DHT_MCU
	11	11:NC32_14
	12	12:NC32_12
	13	13:NC32_13
GND	14	14:GND
ESP32_VIN	15	15:ESP32_VIN

J8
J32R 15pin SOCKET

BUZZ_MCU	1	1:BUZZ_MCU
	2	2:NC32_22
	3	3:NC32_TX0
	4	4:NC32_RX0
LED_B_MCU	5	5:LED_B_MCU
LED_G_MCU	6	6:LED_G_MCU
LED_R_MCU	7	7:LED_R_MCU
	8	8:NC32_0
	9	9:NC32_TX2
	10	10:NC32_RX2
	11	11:NC32_4
	12	12:NC32_2
	13	13:NC32_15
GND	14	14:GND
ESP32_3V3	15	15:ESP32_3V3

MODULE SOCKETS — PIN ORDER PENDING

J2
DHT11 SOCKET

DHT_SENSOR	1	1:DHT_SENSOR
SENSOR_3V3	2	2:SENSOR_3V3
GND	3	3:GND

J3
MQ2 SOCKET

MQ_AO	1	1:MQ_AO
MQ_DO	2	2:MQ_DO
GND	3	3:GND
PSV	4	4:PSV

J4
FLAME SOCKET

SENSOR_3V3	1	1:SENSOR_3V3
GND	2	2:GND
FLAME_DO	3	3:FLAME_DO

J5
BUZZER SOCKET

GND	1	1:GND
BUZZ_INPUT	2	2:BUZZ_INPUT
PSV	3	3:PSV

J6
RGB COMMON CATHODE

GND	1	1:GND
LED_B	2	2:LED_B
LED_G	3	3:LED_G
LED_R	4	4:LED_R

TEST POINTS

TP1
PSV

PSV	1	1:PSV

TP2
SENSOR_3V3

SENSOR_3V3	1	1:SENSOR_3V3

TP3
ESP_3V3

ESP_3V3	1	1:ESP_3V3

TP4
MQ_AO

MQ_AO	1	1:MQ_AO

TP5
ADC32

ADC32	1	1:ADC32

TP6
ADC8266

ADC8266	1	1:ADC8266

TP7
SMOKE_MCU

SMOKE_MCU	1	1:SMOKE_MCU

TP8
FLAME_MCU

FLAME_MCU	1	1:FLAME_MCU

TP9
GND

GND	1	1:GND

J9
J8266L 15pin SOCKET

ADC8266	1	1:ADC8266
	2	2:NC8266_RSV1
	3	3:NC8266_RSV2
	4	4:NC8266_SD3
	5	5:NC8266_SD2
	6	6:NC8266_SD1
	7	7:NC8266_CMD
	8	8:NC8266_SD0
	9	9:NC8266_CLK
GND	10	10:GND
ESP8266_3V3	11	11:ESP8266_3V3
	12	12:NC8266_EN
	13	13:NC8266_RST
GND	14	14:GND
ESP8266_VIN	15	15:ESP8266_VIN

J10
J8266R 15pin SOCKET

LED_B_MCU	1	1:LED_B_MCU
LED_G_MCU	2	2:LED_G_MCU
LED_R_MCU	3	3:LED_R_MCU
	4	4:NC8266_D3
DHT_MCU	5	5:DHT_MCU
ESP8266_3V3	6	6:ESP8266_3V3
GND	7	7:GND
SMOKE_MCU	8	8:SMOKE_MCU
FLAME_MCU	9	9:FLAME_MCU
BUZZ_MCU	10	10:BUZZ_MCU
	11	11:NC8266_D8
	12	12:NC8266_RX
	13	13:NC8266_TX
GND	14	14:GND
ESP8266_3V3	15	15:ESP8266_3V3

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